

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

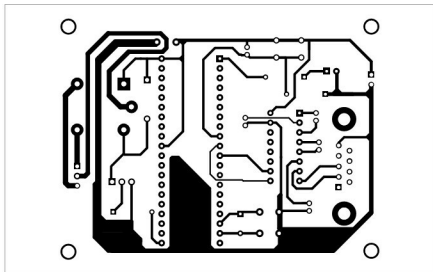


Fig. 9: Actual-size PCB pattern of the sound-operated device-control system

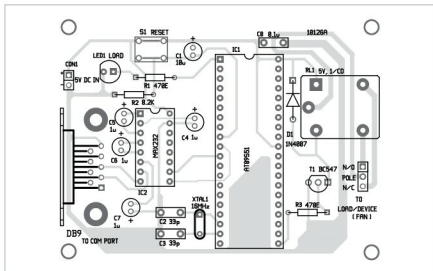


Fig. 10: Component layout of the PCB

this VI, other first two VIs automatically run in the background. Now, give run-time command to the device you want to turn on or off.

Hardware implementation

Fig. 7 shows the circuit diagram of the sound-operated control system. AT89S51 microcontroller is the heart of the system. Output from LabVIEW is given to the controller using USB-to-serial communication cable (DB9 connector). Output command of VI, for instance, 'yes' or 'no', is given to the controller in terms of ASCII.

The controller is programmed

as follows: If received ASCII value from LabVIEW is 'yes,' a relay (RL1) should operate and for 'no' ASCII value, relay should not operate. Device/load is switched on or off through relay RL1.

Author's prototype is shown in Fig. 8. The same can also be implemented on Arduino board. Code for Arduino is also included in this project.

Construction and testing

An actual-size, single-side PCB pattern of the sound-operated device-control system is shown in Fig. 9

PARTS LIST	
Semiconductors:	
IC1	- AT89S51 microcontroller
IC2	- MAX232 driver
T1	- BC547 npn transistor
LED1	- 5mm LED
D1	- 1N4007 rectifier diode
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1, R3	- 470-ohm
R2	- 8.2-kilo-ohm
Capacitors:	
C1	- 10 μ F, 16V electrolytic
C2, C3	- 33pF ceramic disk
C4-C7	- 1 μ F, 16V electrolytic
C8	- 0.1 μ F ceramic disk
Miscellaneous:	
CON1	- 2-pin terminal connector
XTAL1	- 16MHz crystal oscillator
S1	- Tactile switch
RL1	- 5V, 1/CO relay
DB9	- 9-pin COM port connector
	- 5V DC power supply

and its component layout in Fig. 10. After mounting all components on the PCB, connect the serial cable from DB9 connector to COM port of your PC. Also, connect the load across RL1.

Run LabVIEW program. First, select appropriate COM port from LabVIEW front panel and record your voice by clicking on the wav file creating VI and sub VI as shown in Figs 2 and 3.

Run-time voice command is compared with the recorded voice command and output is generated as 'Y' if voice command matches and 'N' if voice command does not match. If 'Y' signal is received in AT89S51, RL1 is energised and the load is turned on. Make sure that the threshold value in cross correlation.vi is 1E-8. **EFY**

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